

# Probabilistic Credit Spread Forecasting:

# Embedding Empirical Distribution Structure into Machine Learning Architectures

By: Madhurima & ShiHan

**How well do Normal and Student-(t) models describe the distribution and tail risk of six-month credit-spread changes?**

**How accurately do their simulated prediction intervals capture realized changes?**



# Data Definition

Source: Historical U.S. High-Yield Option-Adjusted Spread (HY OAS) and U.S. macroeconomic indicators.

| | |
|---|---|
| **HY_OAS** | High-yield option-adjusted spread over Treasuries, percentage points |
| **Fed_Funds_Rate_%** | Federal funds rate, in percent |
| **Unemployment_Rate_%** | US unemployment rate, in percent |
| **Consumer Price Index for All Urban Consumers** | CPI index level for all urban consumers |
| **Global_Commodity_Index** | Broad global commodity price index |

Training Time period: Jan 2000 - Jun 2022

Test Time period: Jan 2023 - Dec 2025

Links:

1. HY-OAS: https://www.macrotrends.net/3006/high-yield-spread
2. U.S. Macroeconomic Indicators: https://fred.stlouisfed.org/

# Feature Engineering

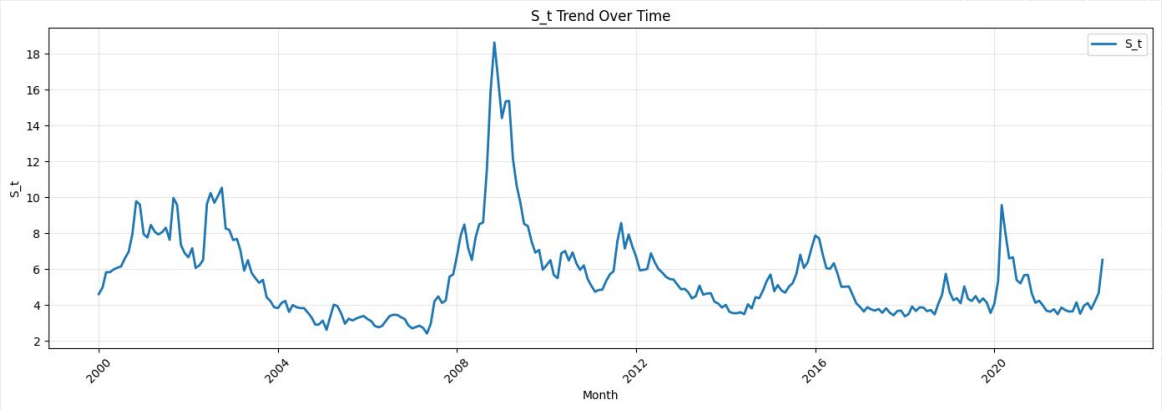
| | |
|---|---|
| **S_t** | Month-end HY OAS, in percentage points |
| **S_t - S_{t-1}** | One-month change in the spread |
| **S_{t+6}** | 6 month shift in credit spread |
| **CPI_yoy** | Year-on-year % change in CPI |
| **GCI_yoy** | Year-on-year % change in the commodity index |
| **Fed_Rate%_L1**<br>**Unemp_Rate%_L1**<br>**CPI_yoy_L1**<br>**GCI_yoy_L1** | Macro predictors lagged one month, to avoid look-ahead |

**Target Variable:**
**Six-month change in the spread**

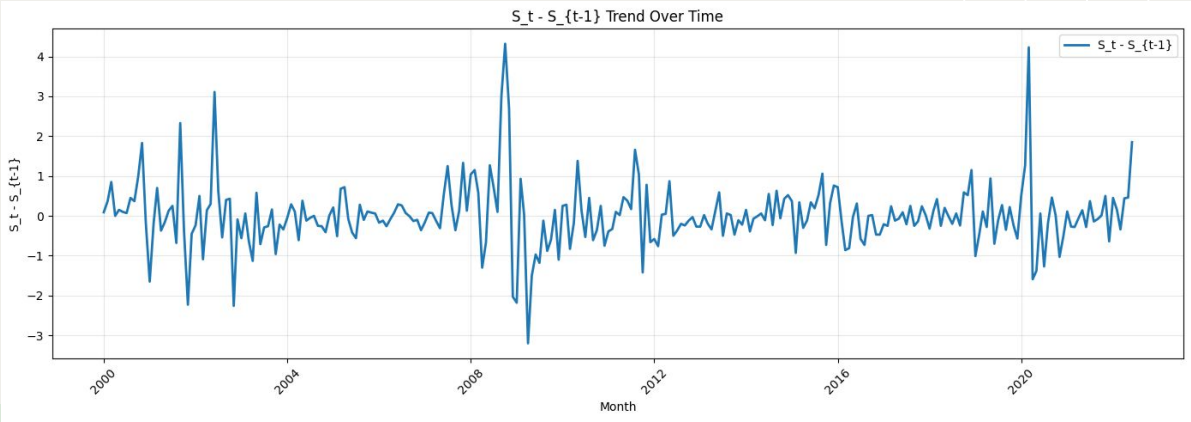
$$\Delta S_{t+6} = S_{t+6} - S_t$$

# Understanding Credit-Spread Movements: Line Graph

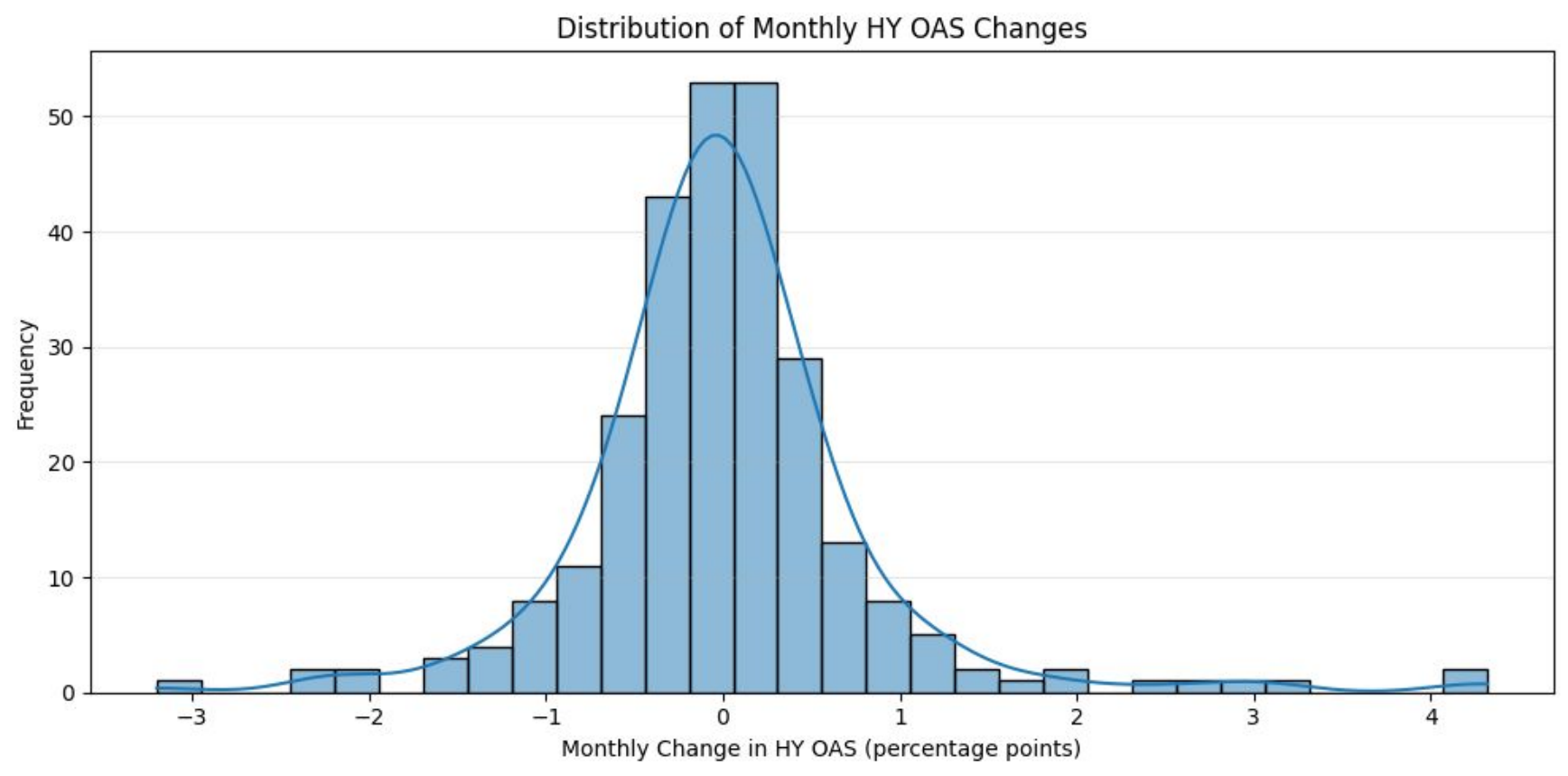
**Spread Level over Time:**




**One-Month Change:**



# Understanding Credit-Spread Movements: Histogram



# Method #1: Comparing Normal and Student-t Distributions

## Normal

$$X \sim N(\mu, \sigma^2)$$

- Describes data using its mean and standard deviation.
- Has relatively thin tails.
- Extreme observations receive very low probability.

## Student-t

$$X \sim t_{\nu}(\mu, \sigma)$$

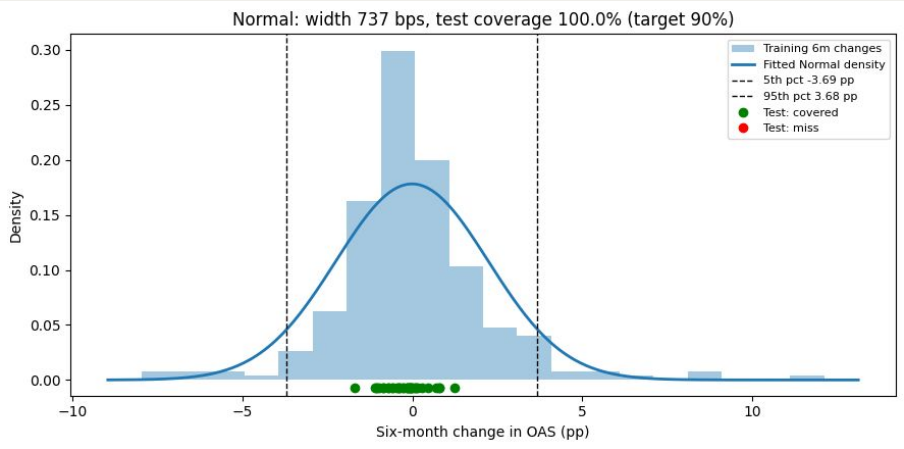
- Has an additional degrees-of-freedom parameter.
- Can accommodate heavier tails.
- Smaller df generally means heavier tails.

## Estimation Method:

1. Fit both distributions using Maximum Likelihood Estimation (MLE).
2. Compare the fitted distributions against the empirical data.
3. Use Akaike Information Criterion (AIC) and Q-Q plots to evaluate model fit.

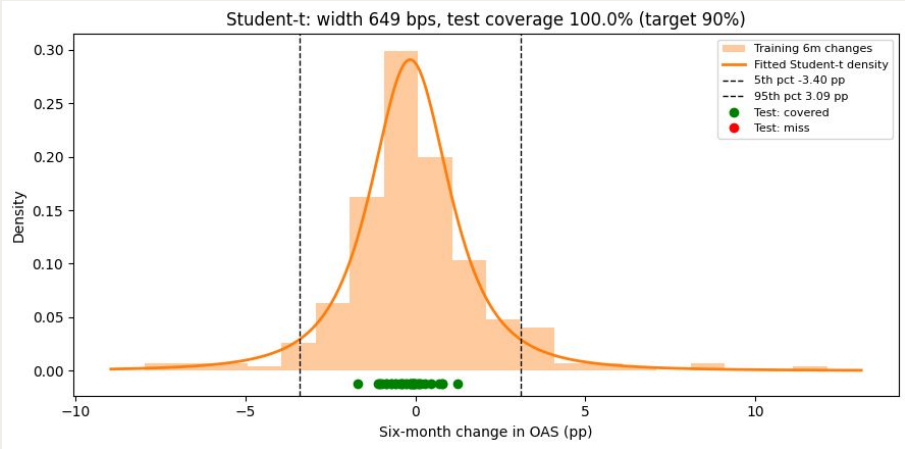
# 6-Monthly Changes: Histogram

## Normal




AIC: 1205.5

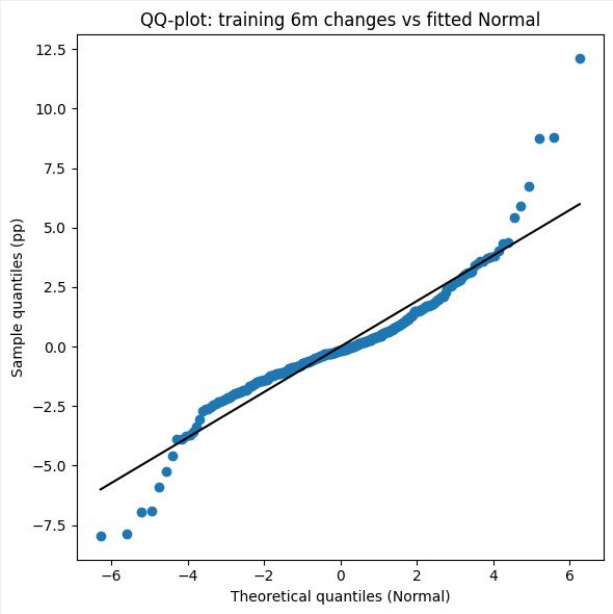
## Student-t

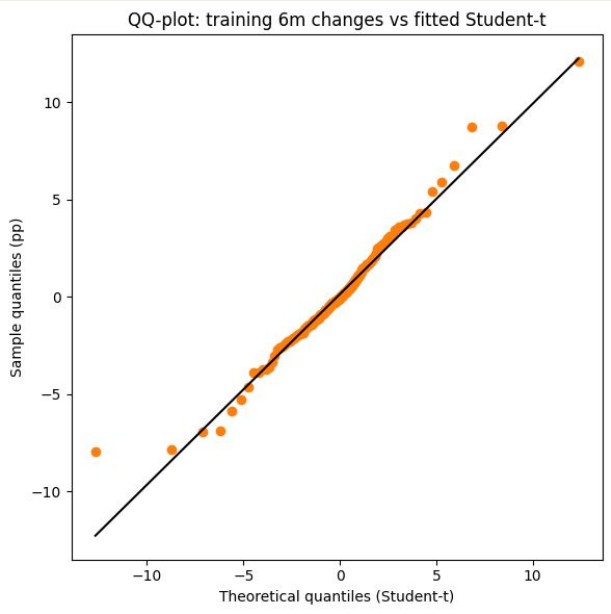



AIC: 1130.7 DF: 2.39

**Fit:** The Student-t matches the peak and the fat tails better, with an AIC about 75 lower (1130.7 vs 1205.5).
**Coverage:** Both bands cover all 36 test points, so coverage can't separate the models. The bands are too wide for the calm 2023–25 window.

# 6-Monthly Changes: QQ-Plot




- The Normal Q-Q plot shows deviations, particularly in the tails.
- The Student-t Q-Q plot follows the empirical quantiles more closely through much of the distribution.
- Both models still struggle with the most extreme observations.

# Method #2: Six-Month Monte Carlo Simulation

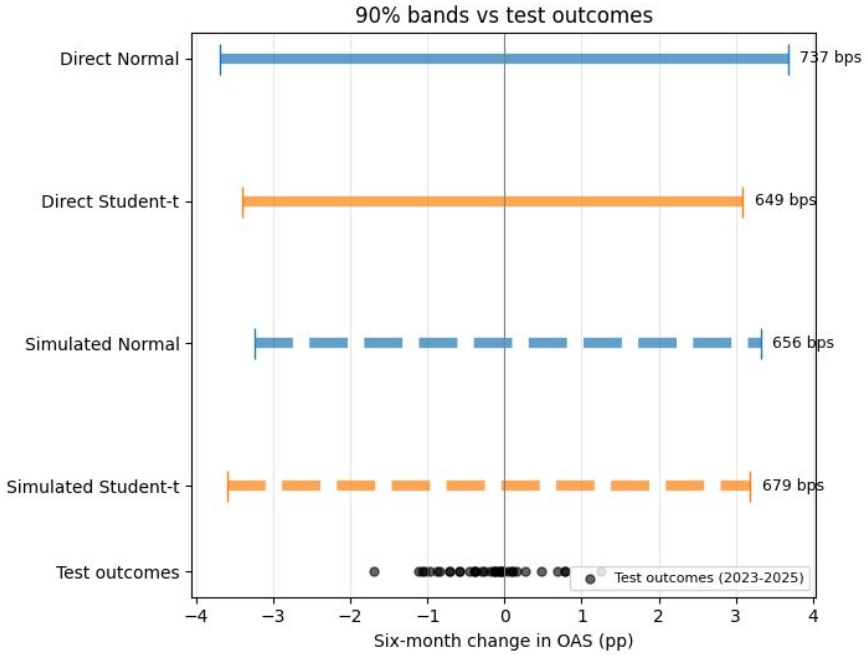
Objective: Estimate the distribution of six-month credit-spread changes by simulating monthly movements.

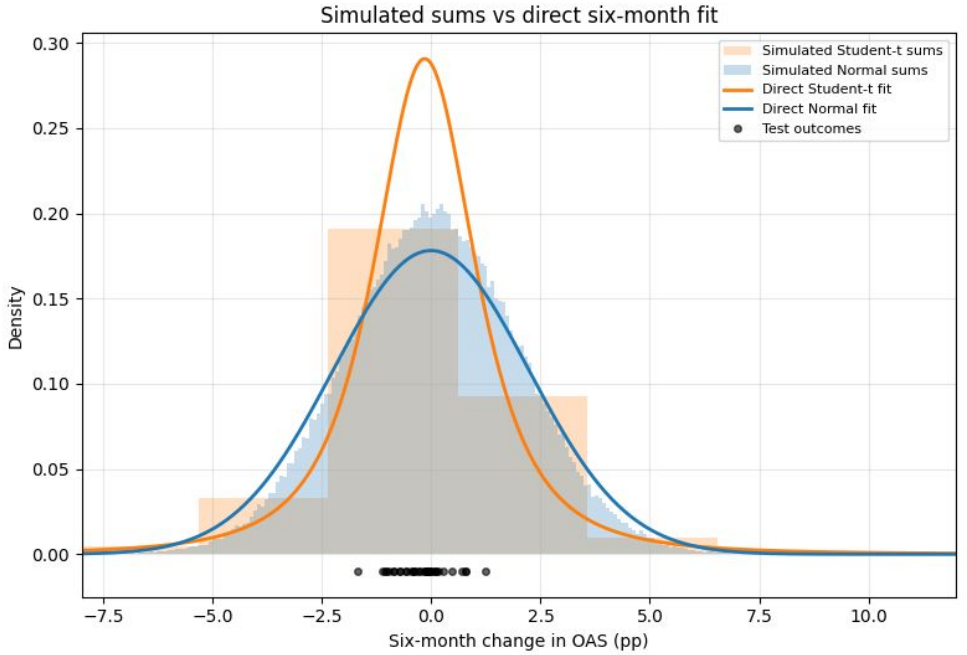
## Methodology

1. Fit Normal and Student-t distributions to historical monthly changes.
2. Generate 200,000 simulated paths.
3. Each path contains six monthly changes.
4. Sum the six monthly changes to obtain one simulated six-month change.
5. Repeat for both distributions.
6. Extract the 5th and 95th percentiles to construct 90% prediction intervals.

$$\Delta S_{t+6} = \sum_{j=1}^{6} \Delta S_{t+j}$$

# Monte-Carlo Simulation Results



# Six-Month Prediction Intervals

| Method | 90% interval (pp) | Width (bps) | Test coverage |
|---|---|---|---|
| Direct Normal | [-3.691, 3.675] | 737 | 100% |
| Direct Student-t | [-3.399, 3.093] | 649 | 100% |
| Simulated Normal | [-3.237, 3.324] | 656 | 100% |
| Simulated Student-t | [-3.604, 3.190] | 679 | 100% |



## Interpretation

- All four intervals captured all 36 test observations.
- The Student-t direct fit produced a narrower interval than the direct Normal fit.
- The simulated intervals are based on six monthly draws rather than directly fitting six-month changes.
- All intervals are relatively wide compared with the realised test-period movements.

Additional Analysis

# Can Macroeconomic Variables Improve Prediction?

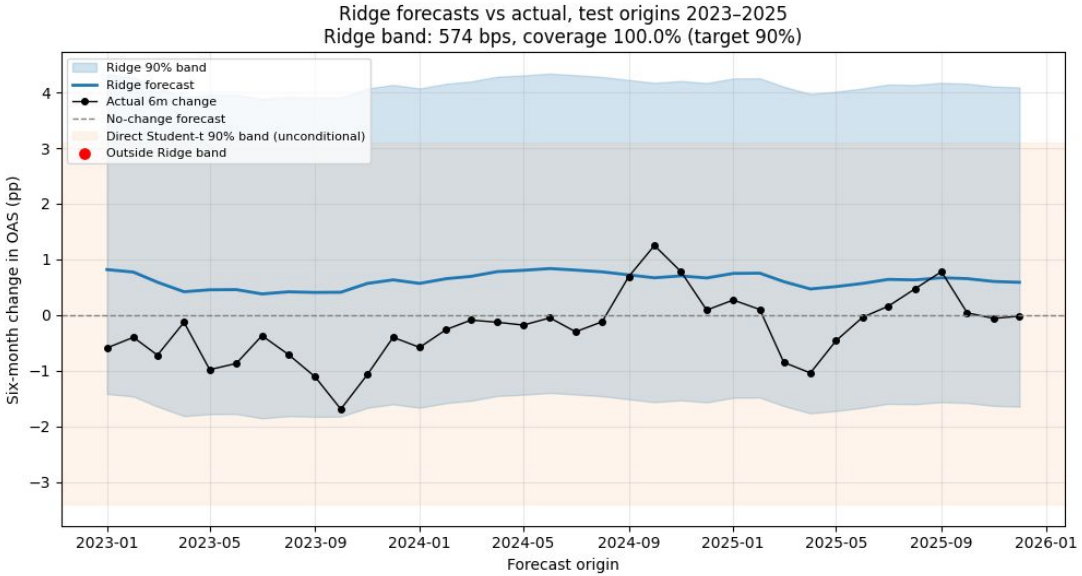
Model Inputs:

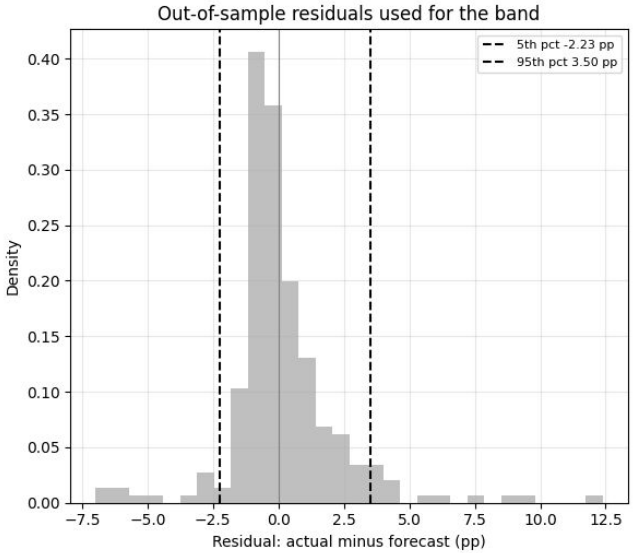
- Current HY OAS.
- Previous one-month spread change.
- Lagged GCI year-on-year change.
- Lagged federal funds rate.
- Lagged unemployment rate.
- Lagged CPI year-on-year inflation.



# Ridge Regression

MAE: Ridge regression - 0.9047 pp | Naive (no-change) - 0.4958 pp

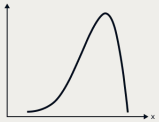


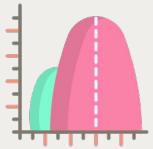

Key Findings:

- Ridge regression performed worse than the simple no-change benchmark.
- Its average forecast was approximately +0.62 pp, while the average realised change was approximately -0.24 pp.
- The model exhibited an upward forecasting bias.
- Its 90% residual prediction interval had a width of approximately 574 bps and covered all 36 test observations.

# Conclusion



Credit-spread changes exhibit heavy-tailed behaviour.



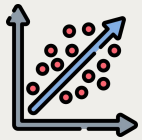
Student-t provides a better fit than Normal.



Relatively calm test period compared with the training period.



All four prediction intervals achieved 100% test coverage, but may be overly conservative.



Recent spread dynamics outperformed lagged macro variables for this test period.

# Limitations



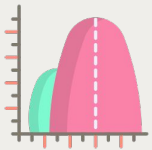
Monte Carlo assumes monthly changes are independent and identically distributed.



Variance ratio of 1.247 suggests the independence assumption may be imperfect.



Only 36 test observations, with overlapping six-month horizons.

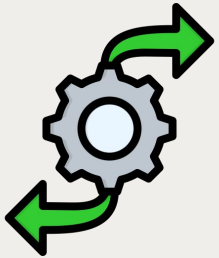


Normal and Student-t distributions may not fully capture asymmetric tail behaviour.

# Future Steps



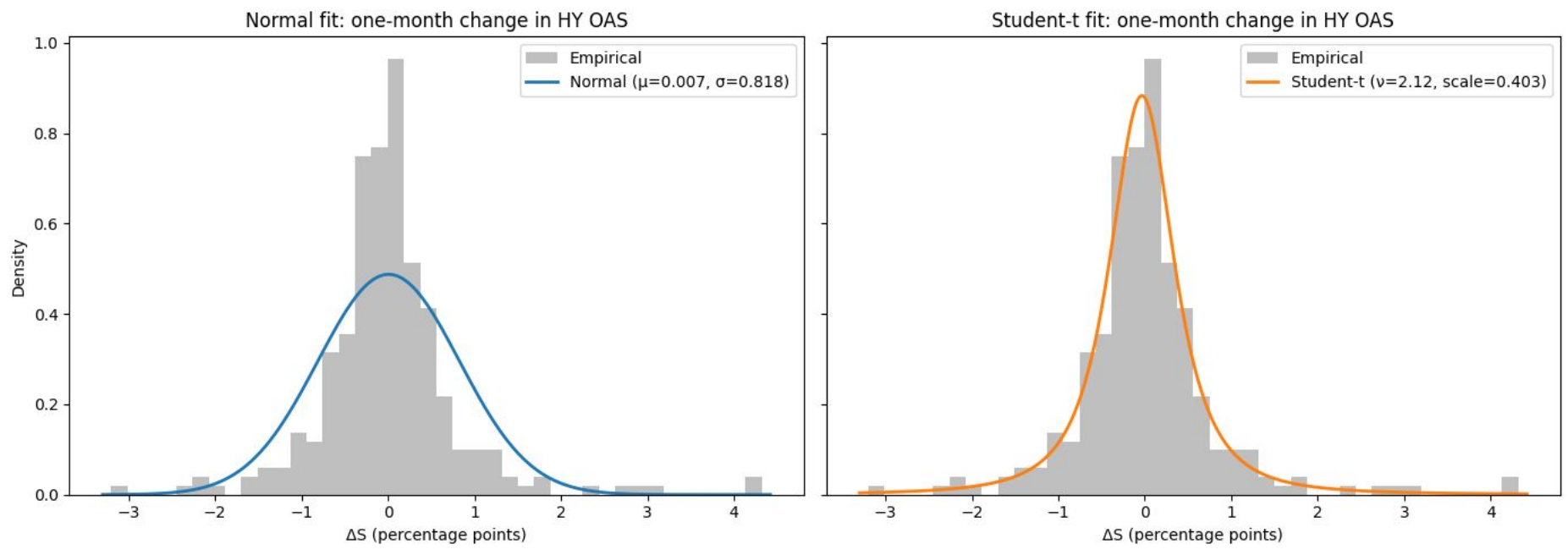
Add autoregressive lags of the monthly change



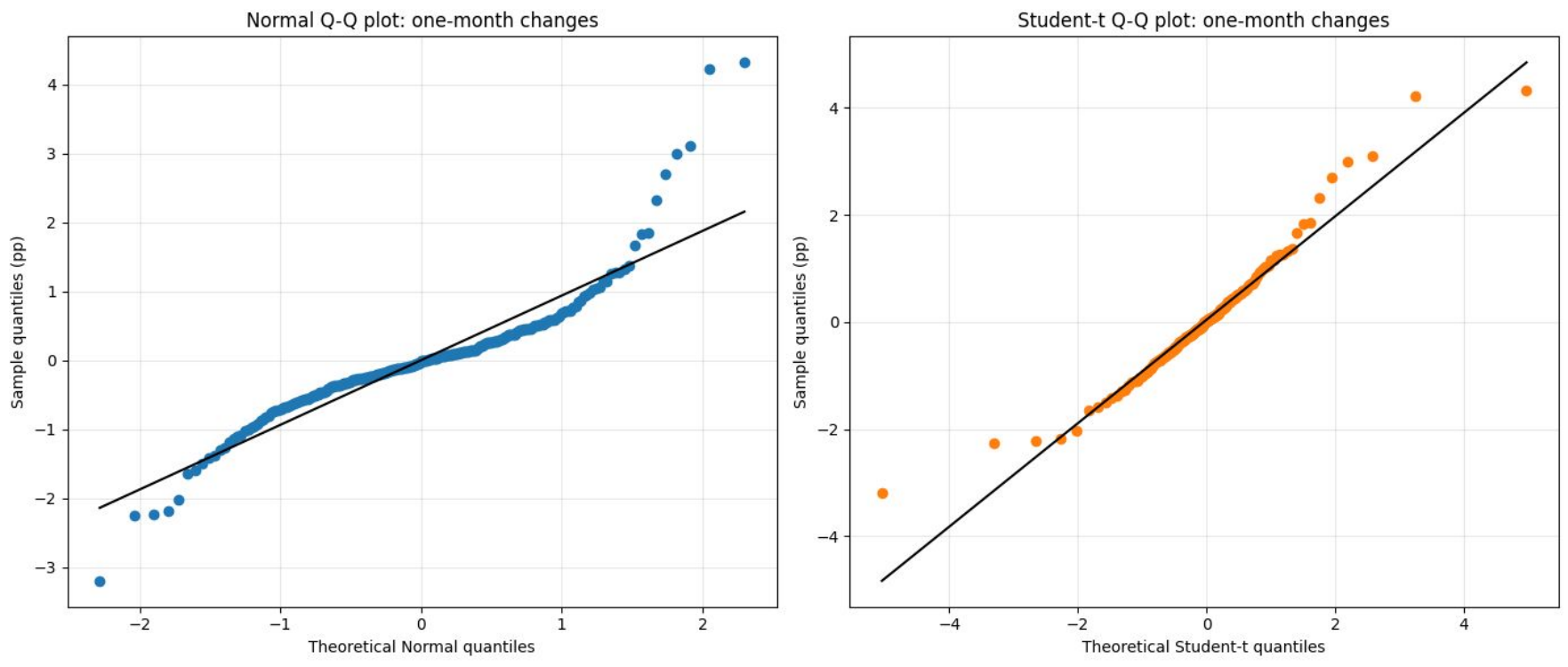
Use a block bootstrap of historical six-month changes instead of simulating independent draws

# Thank You

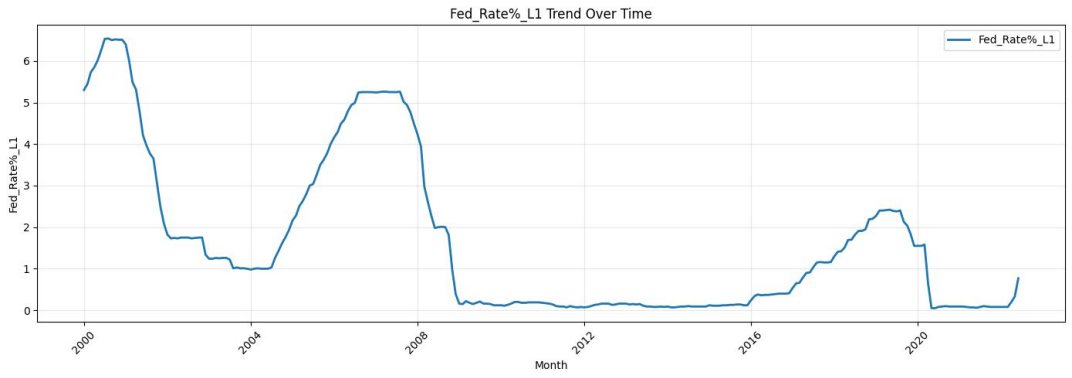
# One-Month Change Distribution

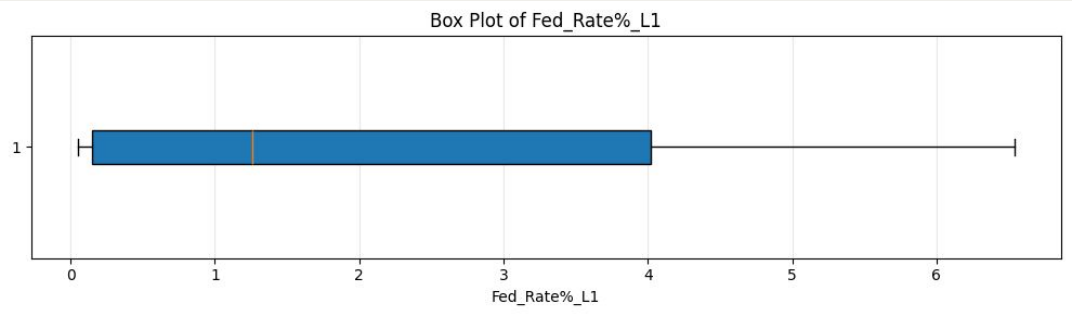


# One-Month Change QQ-Plot

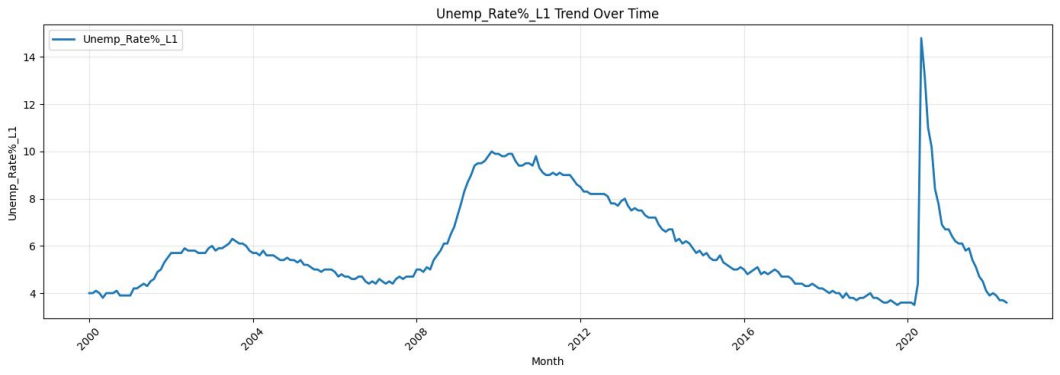


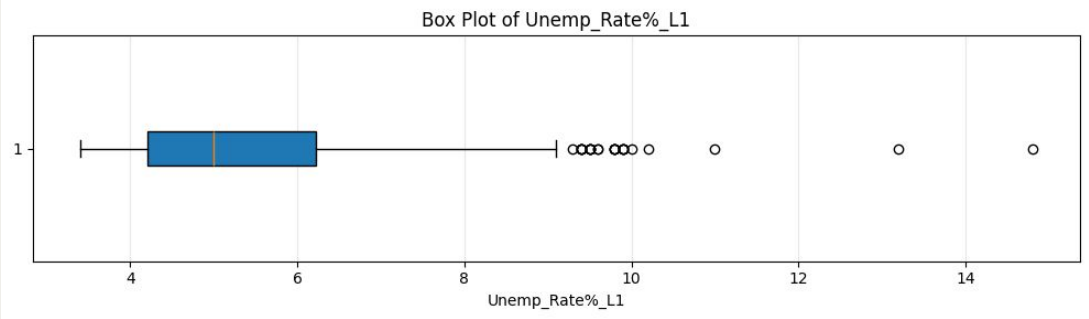
# Fed_Rate%


Fed_Rate%_L1 Trend Over Time
Fed_Rate%_L1
Fed_Rate%_L1
Month
0
1
2
3
4
5
6
2000
2004
2008
2012
2016
2020

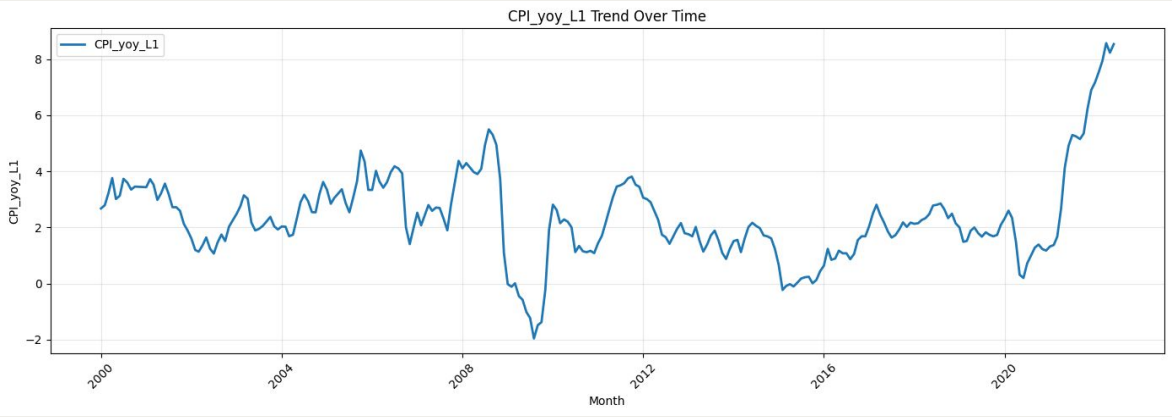

Box Plot of Fed_Rate%_L1
1
0
1
2
3
4
5
6
Fed_Rate%_L1

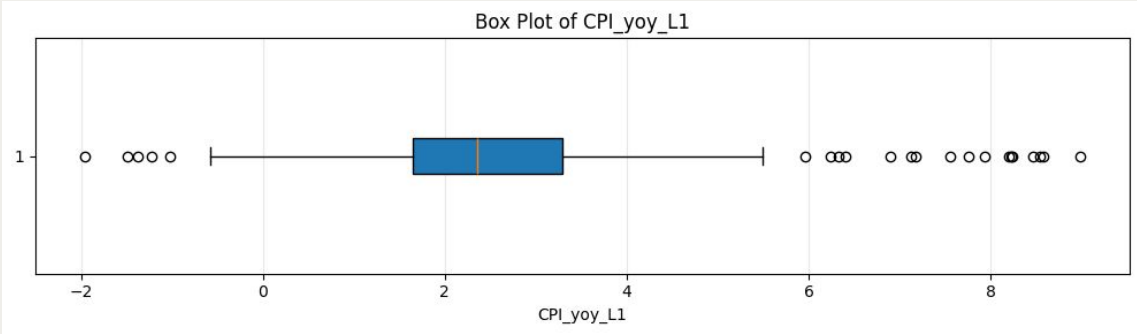
# Unemp_Rate%


Unemp_Rate%_L1 Trend Over Time
Unemp_Rate%_L1
Unemp_Rate%_L1
14
12
10
8
6
4
2000
2004
2008
2012
2016
2020
Month

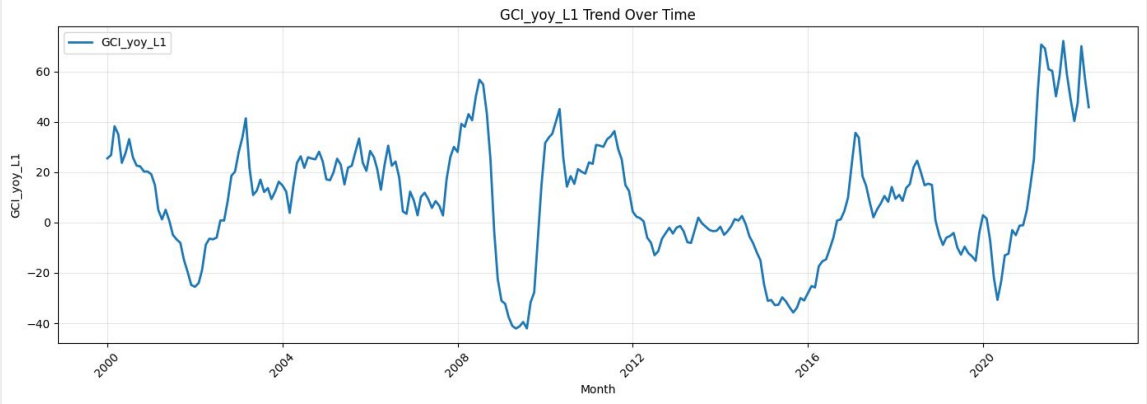

Box Plot of Unemp_Rate%_L1
1
4
6
8
10
12
14
Unemp_Rate%_L1

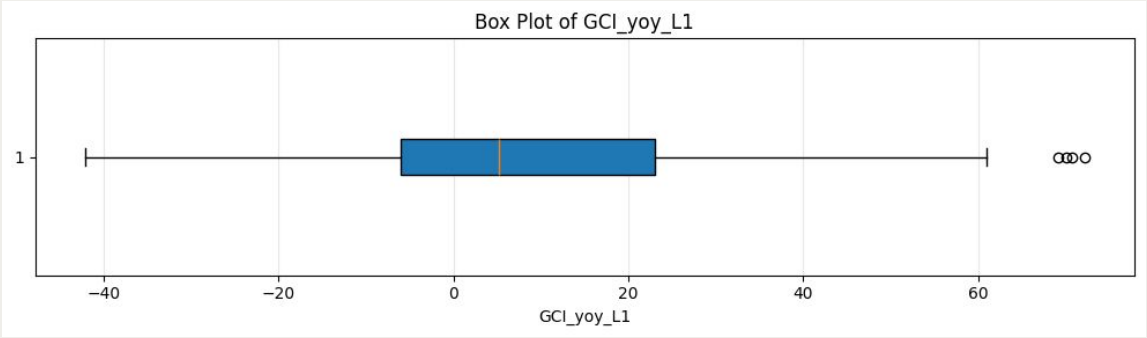
# CPI


CPI_yoy_L1 Trend Over Time
CPI_yoy_L1
CPI_yoy_L1
Month
8
6
4
2
0
−2
2000
2004
2008
2012
2016
2020

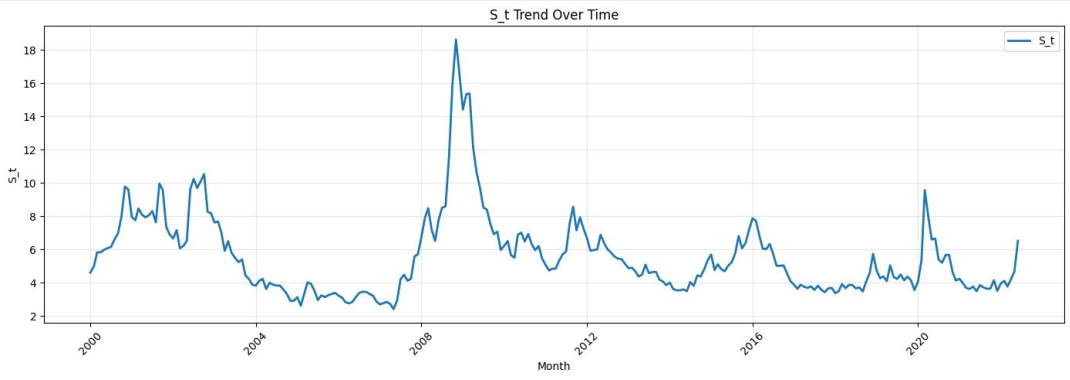

Box Plot of CPI_yoy_L1
1
−2
0
2
4
6
8
CPI_yoy_L1

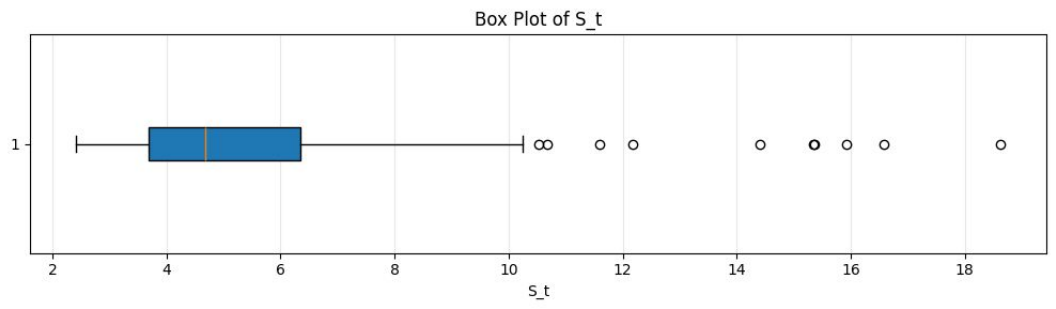
# GCI


GCI_yoy_L1 Trend Over Time
GCI_yoy_L1
60
40
20
0
−20
−40
GCI_yoy_L1
2000
2004
2008
2012
2016
2020
Month

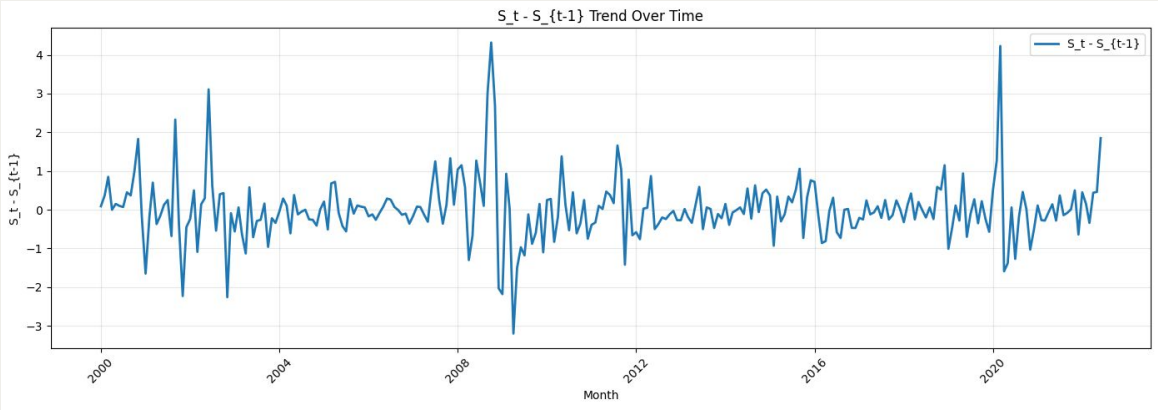

Box Plot of GCI_yoy_L1
1
−40
−20
0
20
40
60
GCI_yoy_L1

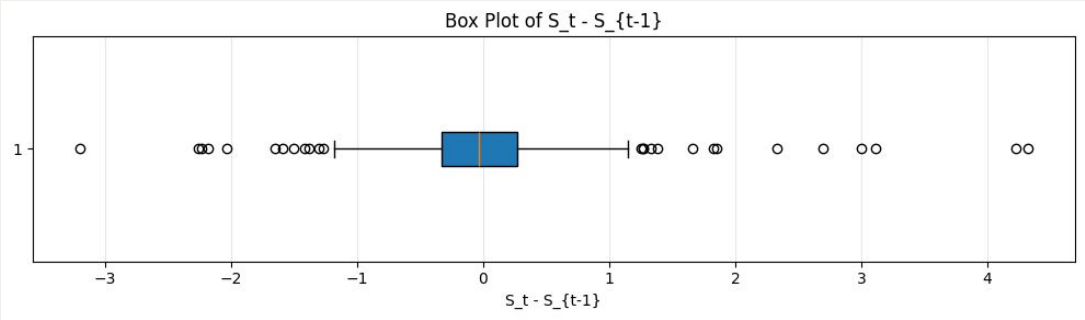
# S_t


S_t Trend Over Time
S_t
18
16
14
12
10
8
6
4
2
S_t
2000
2004
2008
2012
2016
2020
Month

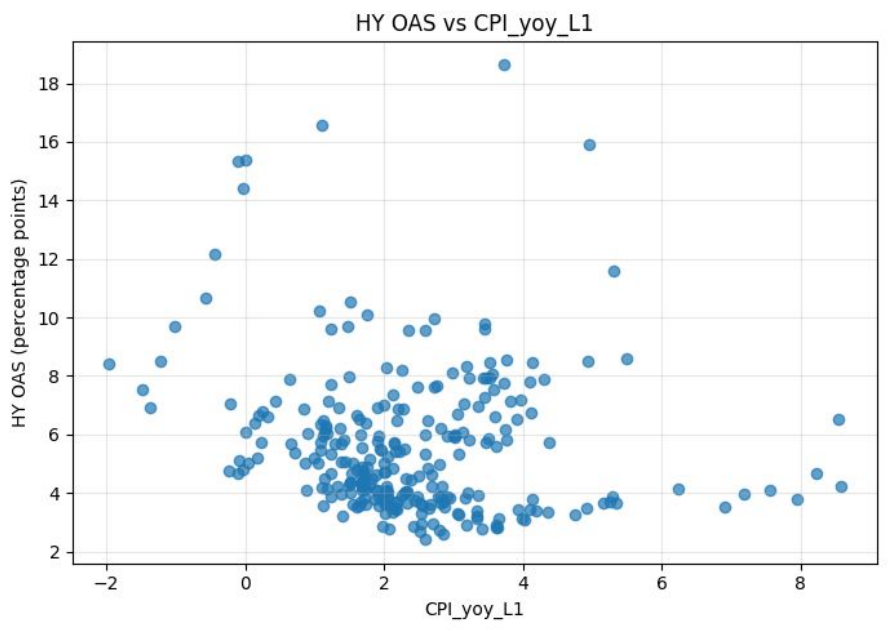

Box Plot of S_t
1
2
4
6
8
10
12
14
16
18
S_t

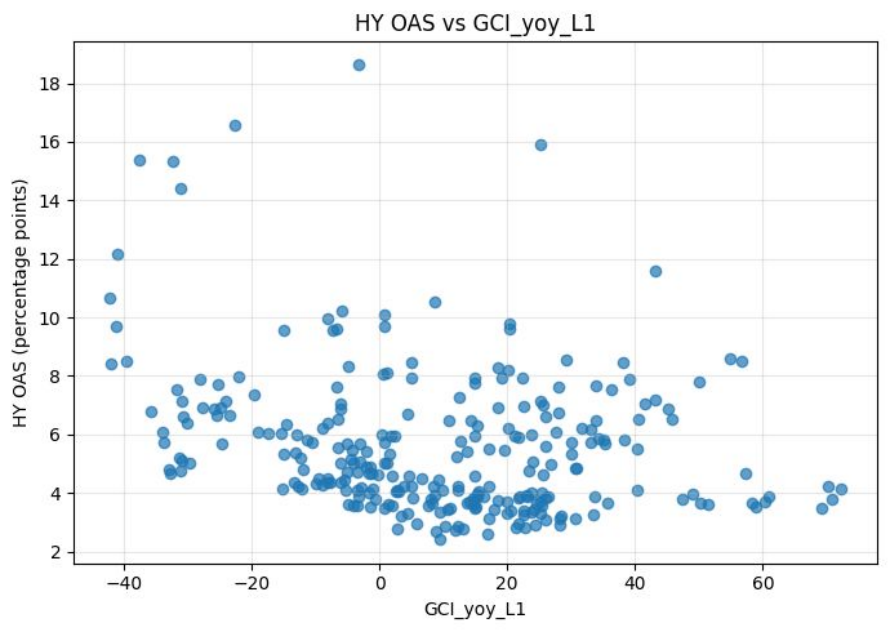
# S_t - S_{t-1}


S_t - S_{t-1} Trend Over Time
S_t - S_{t-1}
4
3
2
1
0
−1
−2
−3
2000
2004
2008
2012
2016
2020
Month

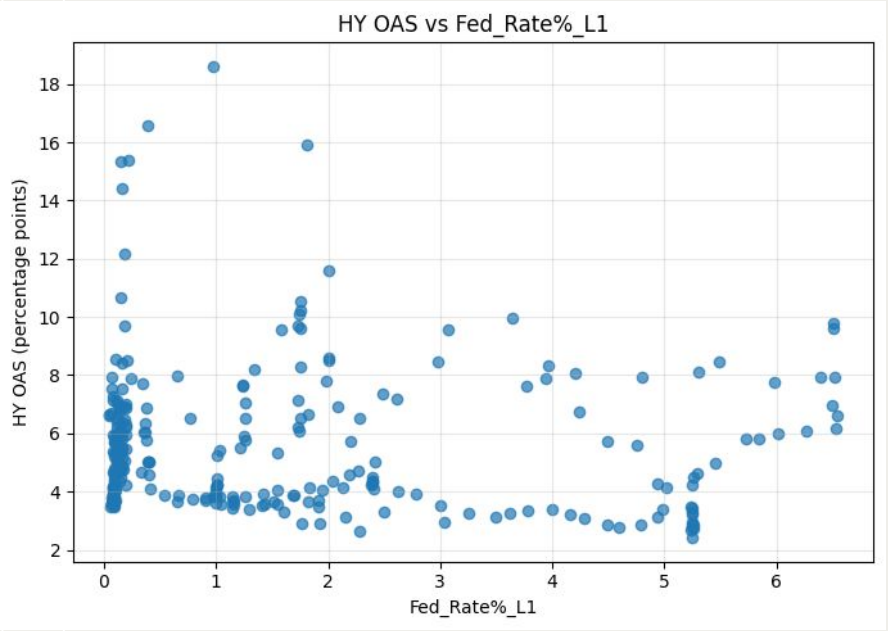

Box Plot of S_t - S_{t-1}
1
−3
−2
−1
0
1
2
3
4
S_t - S_{t-1}

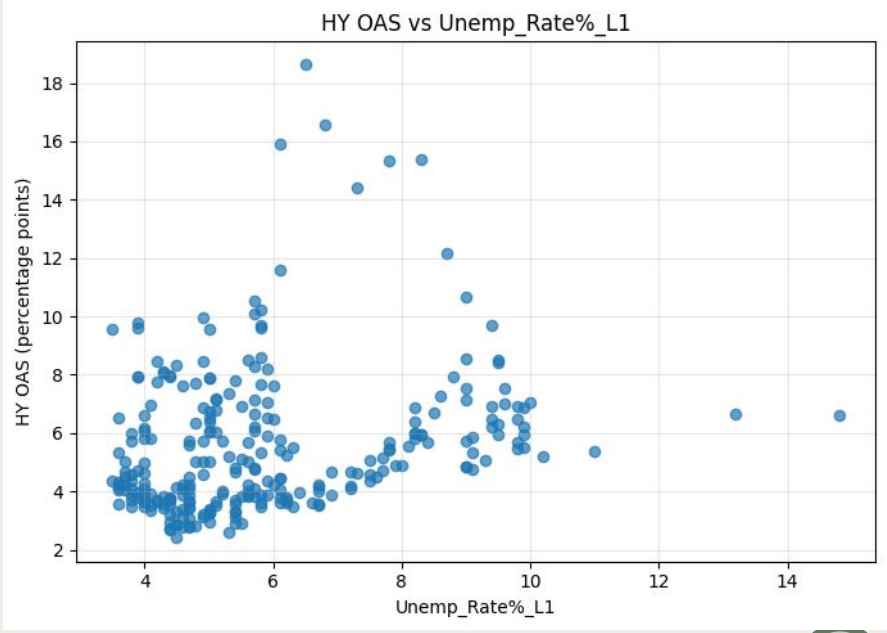
# Scatter Plot



# Scatter Plot



# Correlation Matrix

